ABDERRAHMAN EL FARCHOUNI

Affiliations: International Water Research Institute, Mohammed VI Polytechnic University, Morocco
Department of Agriculture, Food, Environment and Forestry (DAGRI), University of Florence, Florence, Italy
Address: Benguerir, Morocco
Email: elfarchouniabderrahman7@gmail.com | Abderrahman.ELFARCHO@um6p.ma
Phone: +212 644362829
LinkedIn: http://www.linkedin.com/in/abderrahman-el-farchouni
Website: https://elfarchouni.github.io

SUMMARY

PhD candidate in Hydrology, isotopes, hydrochemistry and geophysics at University Mohammed VI Polytechnique (UM6P - IWRI) and Università degli Studi di Firenze (UNIFI - DAGRI) through a cotutelle program. My research focuses on groundwater recharge and water resources management, combining hydrological modeling, remote sensing, geophysics, hydrochemistry, isotopes, and machine learning to improve hydrogeological conceptual models in semi-arid regions. I am experienced in both fieldwork and data analysis, with the ability to design multidisciplinary research frameworks that support sustainable water governance.

EDUCATION

PhD — **Oct 2023 - Present**

- Research focus: Underground hydraulic aquifer systems of the piedmont of the Atlas of Marrakech : hydrogeophysical assessment and modeling

Master — **2021-2023**

Faculty of Sciences, Kenitra

- Master Hydroinformatic and Hydrosystems Management

Bachelor in Applied Geosciences — **2017-2021**

Faculty of Science and Technology, Tangier – Morocco

- Applied Geosciences

DEUST — **2017-2020**

Faculty of Science and Technology, Tangier – Morocco

- Foundation degree (Bac+2) providing multidisciplinary scientific background.

Baccalaureate — **2020**

- Baccalaureate in Mathematics A

Baccalaureate — **2017**

- Life and Earth Sciences

RESEARCH EXPERIENCE

PhD Researcher – Hydrology, hydrochemistry, isotopes and Hydrogeophysics — **Oct 2023 - Present**

Mohammed VI Polytechnic University (UM6P), International Water Research Institute (IWRI)

- Conducting research on groundwater recharge and conceptual modeling of the Haouz–Mejjate aquifer system (Atlas of Marrakech).
- Integration of hydrochemistry, stable isotopes, geophysics, and remote sensing to assess recharge dynamics in semi-arid regions.
- Multidisciplinary framework combining field campaigns, data analysis, and hydrological modeling.
- Integrated Hydrological Modelling

Fieldwork and Data Collection:

- **Geophysical surveys:** Electrical Resistivity Tomography (ERT), Magnetic Resonance Sounding (MRS), seismic surveys, and EM38.
- **Hydrogeological monitoring:** Piezometric network measurement and data analysis.
- **Water sampling:** Collection and analysis of isotopic (δ^2H, $\delta^{18}O$) and hydrochemical samples.
- **Aquifer characterization:** Execution and interpretation of pumping tests.
- **Socio-economic surveys:** Design and implementation of farmer questionnaires on irrigation practices, groundwater recharge, and water management strategies.

PUBLICATIONS

Peer-Reviewed Journal Articles:

[1] El Farchouni, A., Hadri, A., Fakır, Y., Ouarani, M., Azaroual, M., & Kchikach, A.. "Mapping groundwater recharge potential zones in a semi-arid, anthropogenically modified mountainous basin." Scientific African, 30(October), e03025. https://doi.org/10.1016/j.sciaf.2025.e03025, 2025.

[2] El Mezouary, L., Elfarchouni, A., Hadri, A., Hakim, M., Fakır, Y., Bouchaou, L., & Chehbouni, A.. "Geostatistical modeling of subsurface heterogeneity in the Al-Haouz-Mejjate aquifer system, Morocco: A T-PROGS approach for enhanced hydrogeological characterization." Scientific African, 31, e03163. https://doi.org/10.1016/j.sciaf.2025.e03163, 2026.

[3] El Mezouary, L., Kharrou, M. H., Hadri, A., Elfarchouni, A., Fakir, Y., Bouchaou, L., & Chehbouni, A.. "Advanced hydrogeological modeling to reduce uncertainty in groundwater recharge estimation in semi-arid aquifers: Case of Al-Haouz-Mejjate aquifer in Morocco." Chemosphere, 392, 144743, 2025.

[4] El Mezouary, L., Hadri, A., Kharrou, M. H., Fakır, Y., Elfarchouni, A., Bouchaou, L., & Chehbouni, A.. "Contribution to advancing aquifer geometric mapping using machine learning and deep learning techniques: a case study of the AL Haouz-Mejjate aquifer, Marrakech, Morocco." Applied Water Science, 14(5).

Conference Proceedings

[5] El Mezouary, L., Hadri, A., Kharrou, M.H., Fakir, Y., Elfarchouni, A., Bouchaou, L., Chehbouni, A.. "Machine Learning and Deep Learning Guided Assessment of Groundwater Reservoir Hydrodynamic Parameters: A Case Study of The El Haouz Aquifer." 4th International GIRE3D Congress on Participatory and Integrated Management of Water Resources in Arid Zones, Laayoune, Morocco, Nov 2023.

[6] El Farchouni, A., et al.. "Mapping Groundwater Recharge Potential Zones Using Remote Sensing, GIS, and AHP Techniques: A Case Study of the Tensift Basin, Morocco." IAHR Africa Congress, Dec 9–12, 2024, 2024.

[7] El Farchouni, A., Hadri, A., Castelli, G., Fakir, Y., Bresci, E., Ouarani, M., & Kchikach, A.. "Integrating SWAT+ hydrological modelling and remote sensing analysis to estimate surface water balance and groundwater recharge in the High Atlas of Marrakech and the Haouz plain (Morocco)." EGU General Assembly 2026, Vienna, Austria, May 3–8, 2026 (oral presentation, session HS8.2.13, abstract EGU26-21149).

[8] El Farchouni, A., et al.. "Integrating SWAT+ hydrological modelling and remote sensing to estimate surface water balance and groundwater recharge in the High Atlas of Marrakech and the Haouz plain." 5th International Conference on African Rivers (ICAR 2026), Rabat, Morocco, Jul 6–11, 2026 (oral communication).

TECHNICAL SKILLS

Remote Sensing & GIS:

- **Google Earth Engine (GEE)** for large-scale geospatial analysis (land use/land cover, NDVI/NDWI, evapotranspiration modeling, groundwater recharge mapping).
- **ArcGIS, QGIS** for spatial analysis, interpolation, and cartography.
- Satellite data processing (Sentinel-2, Landsat, CHIRPS, WaPOR, TerraClimate, SMAP, GPM etc).

Hydrological & Hydrogeological Modeling:

- **SWAT/SWAT+ and MODFLOW** for integrated surface–groundwater modeling
- **GMS:** Groundwater Flow modeling

Geophysics & Field Methods:

- Electrical Resistivity Tomography (ERT), Magnetic Resonance Sounding (MRS), seismic surveys, EM38.
- Pumping tests, piezometric monitoring, isotopes (δ^2H, $\delta^{18}O$) and hydrochemistry sampling.
- Geophysics software: Prodiviner, Res2DInv

Programming & Data Science:

- **Python** (pandas, scikit-learn, TensorFlow, Xarray, matplotlib etc).
- **Machine learning & deep learning** (KNN, Random Forest, LSTM, GAIN, PINN)
- **R for statistical and hydrogeochemical analysis** (hydrogeochemistry diagrams, PCA, clustering).
- **SQL & SQLite Studio** for database management.

Other Tools:

- **Adobe Illustrator** for scientific figure editing and publication-quality graphics.
- **Microsoft Office** Suite (Excel, Word, PowerPoint).
- **LaTeX** for academic writing and reports.

CONFERENCES WORKSHOPS & TRAINING

- **EGU General Assembly 2026, Vienna, Austria:**
  - Oral presentation: Integrating SWAT+ hydrological modelling and remote sensing analysis to estimate surface water balance and groundwater recharge in the High Atlas of Marrakech and the Haouz plain (Morocco), session HS8.2.13 (May 3–8, 2026).
- **5th International Conference on African Rivers (ICAR 2026), Rabat, Morocco:**
  - Oral communication: Integrating SWAT+ hydrological modelling and remote sensing to estimate surface water balance and groundwater recharge in the High Atlas of Marrakech and the Haouz plain (Jul 6–11, 2026).
  - Post-conference training: Machine Learning for Hydrological Impacts Mapping (Jul 10–11, 2026).
  - Post-conference training: Deep Learning for Hydrological Time Series Prediction (Jul 10–11, 2026).
- **IAHR Africa Congress:**
  - Mapping Groundwater Recharge Potential Zones Using Remote Sensing, GIS, and AHP Techniques: A Case Study of the Tensift Basin, Morocco (Dec 9–12, 2024).
- **Discover Scopus AI (May 15, 2024).**
- **Workshop on Isotopic Tracers in Hydrology and Earth Sciences** (May 28–29, 2024).
- **Karst and Cave Sustainable Management Summer Course** (Jul 29 – Aug 9, 2024).
- **Enhancing Water Productivity with WaPOR: Using WaPLUGIN in QGIS** (Dec 12, 2024).
- **Mathematical Models and Artificial Intelligence for Solving Groundwater Flow Problems** (Jan 6–7, 2025).
- **Maghrebian Colloquium of Applied Geophysics (MCGA-9) – Active participation in the 9th edition, Faculty of Sciences, Ibn Tofail University, Kenitra, Morocco** (Apr 23–25, 2024).
- **SYMPLE: Introduction to Applied Groundwater Flow Modelling with GW Vistas – online training** (Mar 17–19 & Apr 7–10, 2025).
- **SYMPLE: Understand MODFLOW** (May 6–27, 2025).

LANGUAGES

- **Arabic (Native)**
- **French (Fluent)**
- **English (Fluent)**